E-Commerce Sales & Customer Analytics

End-to-End Data Analytics Project — Final Report

Tools: Python, Pandas, NumPy, Google Colab, PostgreSQL, SQL, pgAdmin 4, Microsoft Excel, Power BI

Data scope: 1,000 customers, 5,000 orders, 8,345 order-item records and 41 products.

Data period: 1 January 2025 to 14 October 2026 (based on the uploaded orders dataset).

Project objective

To clean, organize, analyze and visualize e-commerce sales and customer data across Python, SQL, Excel and Power BI, then translate the results into practical business insights.

Prepared from the cleaned CSV datasets and Excel workbook supplied for this project. Values below are calculated from the uploaded data.

1. Executive Summary

The dataset contains 5,000 orders and 8,345 order-item records from 1,000 customers and 41 products. Summed order-item sales amount is ■22.63 million and total recorded profit is ■6.37 million, giving a profit-to-sales ratio of 28.16%. Electronics generates the highest category sales, while Furniture generates the highest category profit. August has the highest monthly sales total and November the lowest. Mumbai is the highest-sales city and Solapur the lowest among the listed cities.

Metric	Result
Total Sales (order-item sales amount)	■22,630,978.14
Total Profit	■6,373,807.52
Profit Margin (profit / sales)	28.16%
Orders	5,000
Customers	1,000
Products	41
Order-item records	8,345
Total quantity	15,867
Delivered orders	2,493 (49.86%)
Returned orders / order return rate	628 / 12.56%

2. Data, Tools and Workflow

Python / Google Colab: used for data preparation and exploratory analysis. **PostgreSQL / pgAdmin 4:** used to store the four tables and run SQL queries including filtering, aggregation, joins, CTEs and window functions. **Excel:** used for lookup-based enrichment, Pivot Tables, charts and KPI summary. **Power BI:** used to import the cleaned datasets, establish table relationships and build an interactive report.

The four datasets are customers, orders, order_items and products. The intended relationship path is customers → orders → order_items, with products → order_items.

3. Data Quality Checks

The uploaded cleaned CSVs contain no missing values and no duplicate rows in the checks performed for this report. The customer, order and product keys used by the relationships also matched across the related tables: no unmatched customer IDs in orders, no unmatched order IDs in order items, and no unmatched product IDs in order items.

Note: These checks describe the supplied cleaned files. They do not independently validate the original source system or business rules behind the synthetic data.

4. Sales and Profit by Category

Electronics leads sales at ■5.47 million. Furniture ranks second in sales but records the highest profit at ■1.69 million. Stationery has the lowest sales and profit among the listed categories.

Category	Sales	Profit
Electronics	■5,469,410.64	■1,631,363.06
Furniture	■5,051,495.29	■1,690,475.50
Home & Kitchen	■3,702,522.81	■1,050,810.69
Fashion	■2,876,417.93	■658,926.53
Footwear	■2,713,983.70	■557,007.11
Books	■1,407,710.61	■425,086.17
Fitness	■1,126,701.25	■271,946.85
Stationery	■282,735.91	■88,191.61

5. Monthly Sales

August has the highest monthly sales total (■2.36 million), while November has the lowest (■1.04 million). Monthly figures are grouped by month name across the available data period, which includes both 2025 and 2026; they are not a single-year month-by-month trend.

Month	Sales
August	■2,355,987.30
March	■2,212,557.11
July	■2,206,418.35
January	■2,177,368.74
May	■2,114,838.68
April	■2,056,392.39
June	■2,035,713.70
September	■1,981,701.24
February	■1,826,203.53
October	■1,576,576.56
December	■1,046,813.31
November	■1,040,407.23

6. City and Customer Segment Performance

Mumbai is the highest-sales city at ■1.97 million. Solapur is the lowest at ■1.24 million. Regular customers account for the largest segment sales total (■8.05 million), followed by New and Premium segments.

City	Sales
Mumbai	■1,967,549.86
Thane	■1,901,410.46
Kolhapur	■1,857,651.43
Amravati	■1,842,300.60
Aurangabad	■1,775,438.91
Jalgaon	■1,656,389.18
Satara	■1,648,689.26
Navi Mumbai	■1,641,541.16
Nashik	■1,616,625.58
Pimpri-Chinchwad	■1,541,902.54
Pune	■1,326,103.51
Nagpur	■1,320,904.61
Sangli	■1,296,856.71
Solapur	■1,237,614.33

Customer segment	Sales
Regular	■8,050,600.93
New	■7,327,124.41
Premium	■7,253,252.80

7. Top Products

Office Chair (P030) is the top product by both total sales and total profit in this dataset. External SSD (P005) is the second-highest profit contributor among the five products listed below.

Product ID	Product	Sales	Profit
P030	Office Chair	■2,074,302.60	■794,700.80
P024	Air Fryer	■1,735,286.39	■568,452.47
P005	External SSD	■1,491,662.34	■570,993.36
P031	Study Table	■1,463,878.77	■461,379.30
P003	Smart Watch	■1,281,884.86	■345,008.51

8. Payment Methods and Order Status

Credit Card is the most frequently recorded payment method (1,033 orders), closely followed by UPI (1,020). Delivered is the largest order-status group with 2,493 orders (49.86%). Returned orders number 628, equivalent to a 12.56% order return rate using all 5,000 orders as the denominator.

Payment method	Order count
Credit Card	1,033
UPI	1,020
Cash on Delivery	991
Debit Card	981
Net Banking	975

Order status	Orders	Sales amount attached to those orders*
Delivered	2,493	■11,220,687.62
Shipped	646	■2,964,798.95
Returned	628	■2,894,016.80
Cancelled	627	■2,779,567.15
Processing	606	■2,771,907.62

*Sales amount shown by order status is the order-item sales amount associated with those orders in the dataset. It should not be treated as realized revenue for cancelled or returned orders without refund/cancellation reconciliation.

9. Key Findings and Recommendations

Findings

• Total sales amount: ■22,630,978.14; total recorded profit: ■6,373,807.52; profit-to-sales ratio: 28.16%.
• Electronics leads category sales; Furniture leads category profit.
• Office Chair is the top product by sales and profit.
• August is the highest-sales month name; November is the lowest in the month-name aggregation.
• Mumbai has the highest city sales; Solapur has the lowest.
• Credit Card and UPI are the two most frequent payment methods, with very similar counts.
• 12.56% of orders are marked Returned; 49.86% are marked Delivered.

Recommendations

1. Investigate return reasons and product-level return patterns before deciding on corrective actions.
2. Review the performance of lower-sales cities such as Solapur and Sangli; assess local demand, delivery coverage and marketing reach before allocating more budget.
3. Maintain availability and monitor margins for Electronics and Furniture, since these categories lead sales and profit respectively.
4. Examine monthly patterns using a year-month field in Power BI so the 2025 and 2026 trends can be compared without combining same-named months across years.
5. Reconcile cancellations and returns with refunds before reporting net realized revenue.

10. Limitations and Interpretation Notes

• The data is synthetic/sample project data and should not be presented as verified results from a real retailer.
• Profit margin is calculated as total recorded profit divided by total order-item sales amount; it is not a net margin unless all relevant costs are included in the supplied profit field.
• The return rate uses returned orders divided by all orders. It is an order-level rate, not an item-level return rate.
• The month summary aggregates month names across two calendar years. Use Year + Month for time-series analysis.
• The sales amounts for Returned and Cancelled statuses are recorded order-item amounts and have not been adjusted for refunds.

Conclusion

This project demonstrates an end-to-end analytics workflow across Python, SQL, Excel and Power BI. It provides a structured view of sales, profit, category and product performance, geography, payment methods and order statuses. The most important follow-up areas are return analysis, accurate year-month comparisons and reconciliation of returned/cancelled sales before estimating realized revenue.